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MINI PROJECT REPORT

ON

Hotel Network Design with Guest and Administration VLANs

BACHELOR OF SCIENCE (HONS.)

In

COMPUTER SCIENCE

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I would like to express my sincere gratitude to my faculty member for providing guidance and support throughout this Computer Networks mini project. This project helped me understand how VLANs, trunk links, router subinterfaces, IP addressing, and Access Control Lists (ACLs) can be used to design a segmented network.

I am also thankful to my institution for providing the opportunity and learning environment to complete this practical work. I acknowledge Cisco Packet Tracer and the networking learning resources that helped me build, configure, and test the proposed network.

Finally, I would like to thank everyone who directly or indirectly supported me in completing this mini project successfully.

ABSTRACT

This project presents a simulated hotel network that separates guest devices from administration devices using Virtual Local Area Networks (VLANs). The network is designed in Cisco Packet Tracer using one Cisco 1941 router, one Cisco 2960 switch, and four PCs. VLAN 10, named GUEST, is assigned to two guest PCs, while VLAN 20, named ADMIN, is assigned to two administration PCs.

A trunk link carries both VLANs between the switch and router, and router-on-a-stick subinterfaces provide the default gateways for the two IP networks. An extended Access Control List (ACL) is applied inbound on the guest router subinterface to deny traffic from the Guest subnet to the Administration subnet.

Connectivity tests verify communication within each VLAN and between PCs and their respective gateways. A Guest-to-Administration ping is expected to be blocked by the configured ACL. The project demonstrates network segmentation and directional access control in a simulated environment. It does not include a production internet connection, wireless access points, DHCP, or NAT.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Hotels provide network access to different groups of users, including guests and administration staff. These users have different network access requirements. Guests generally need access to basic network services, while administration staff may use internal systems for hotel management and other administrative tasks.

If all devices are connected to a single network without proper separation, it becomes more difficult to manage traffic and apply different security policies. Therefore, separating the guest and administration networks is an important part of network design.

A Virtual Local Area Network (VLAN) logically divides a switched network into separate broadcast domains. In this project, VLAN 10 is used for guests and VLAN 20 is used for administration. The router provides gateways for both networks, while an Access Control List restricts traffic from the Guest subnet to the Administration subnet.

The network is designed and tested using Cisco Packet Tracer, which allows network devices and configurations to be simulated in a virtual environment.

1.2 Problem Statement

The problem addressed by this project is how to separate guest devices and administration devices in a small hotel network and prevent guest devices from initiating IP communication to the administration subnet.

Each group should have its own IP subnet and default gateway while retaining basic connectivity within its own group. The network should also demonstrate how VLANs, trunking, routing, and access-control policies can be used to improve network organization and security.

1.3 Aim and Objectives

Aim

To design and simulate a hotel network with separate Guest and Administration VLANs and apply a basic access-control policy using Cisco Packet Tracer.

Objectives

1. To create VLAN 10 (GUEST) and VLAN 20 (ADMIN) on the switch.
2. To assign guest PCs and administration PCs to their respective VLAN access ports.
3. To configure a trunk link between the switch and router to carry VLANs 10 and 20.
4. To configure router-on-a-stick subinterfaces and default gateways for both VLANs.
5. To assign appropriate IP addresses, subnet masks, and default gateways to all PCs.
6. To apply an extended ACL to deny Guest-subnet traffic to the Administration subnet.
7. To test gateway connectivity and communication between PCs in the same VLAN.
8. To verify that Guest-to-Administration traffic is blocked by the configured ACL.

1.4 Scope of the Project

The project covers a simulated network containing four end devices, one Layer 2 switch, and one router. It includes VLAN creation, access-port assignment, IEEE 802.1Q trunking, IPv4 addressing, router subinterfaces, and an extended ACL.

The project focuses on basic network segmentation and access control in a small hotel environment. It is intended for educational purposes and demonstrates fundamental networking concepts.

Wireless networking, internet connectivity, Network Address Translation (NAT), Dynamic Host Configuration Protocol (DHCP), server services, and a restriction on Administration-to-Guest traffic are outside the current implementation. These features can be added in future versions of the project.

CHAPTER 2: SYSTEM REQUIREMENTS AND DESIGN

2.1 Hardware Requirements

The following simulated network devices are used in the project.

2.1 Hardware Requirements

Component	Quantity	Purpose
Cisco 1941 router (simulated)	1	Provides gateway interfaces and routes between IP subnets.
Cisco 2960 switch (simulated)	1	Connects PCs and separates them into VLANs.
PC end devices (simulated)	4	Two guest clients and two administration clients.
Ethernet links (simulated)	5	Connect PCs to the switch and switch to router.

All devices are simulated in Cisco Packet Tracer; physical networking hardware is not required for this implementation.

2.2 Software Requirements

2.2 Software Requirements

Software / technology	Purpose
Cisco Packet Tracer	Builds and tests the simulated network.
Cisco IOS-style CLI	Configures VLANs, trunking, subinterfaces, and ACLs.
IPv4	Provides logical addressing for the two subnets.
IEEE 802.1Q	Tags VLAN traffic across the trunk link.

2.3 System Architecture

The network consists of four PCs connected to a Cisco 2960 switch. PC0 and PC1 belong to VLAN 10 (GUEST), while PC2 and PC3 belong to VLAN 20 (ADMIN).

The switch connects to Router0 through switch port FastEthernet 0/24 and router interface GigabitEthernet 0/0. The switch port is configured as a trunk, allowing traffic from VLANs 10 and 20 to travel over the same physical link.

The router uses two subinterfaces:

- GigabitEthernet 0/0.10: Handles VLAN 10 and uses IP address 192.168.10.1.
- GigabitEthernet 0/0.20: Handles VLAN 20 and uses IP address 192.168.20.1.

Each subinterface acts as the default gateway for its respective VLAN. An extended ACL is applied inbound on the Guest subinterface to deny traffic from the Guest subnet to the Administration subnet.

Network Topology Diagram

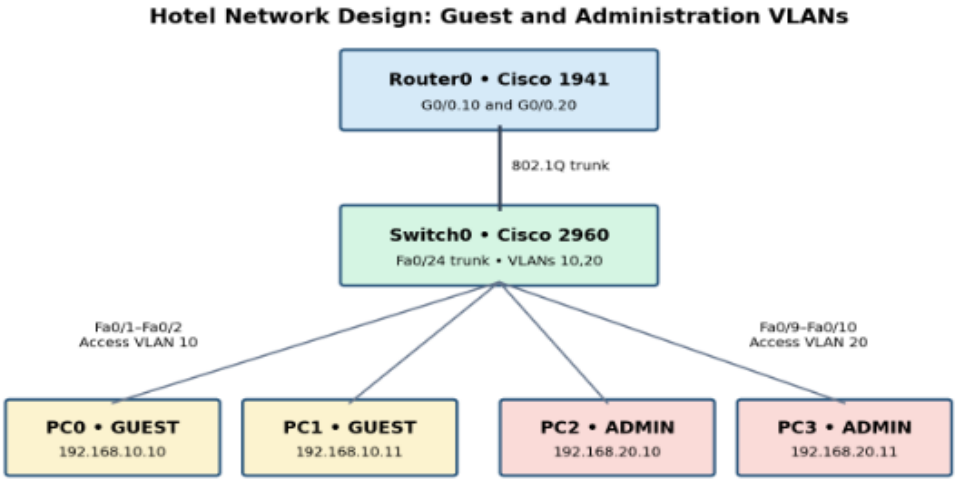
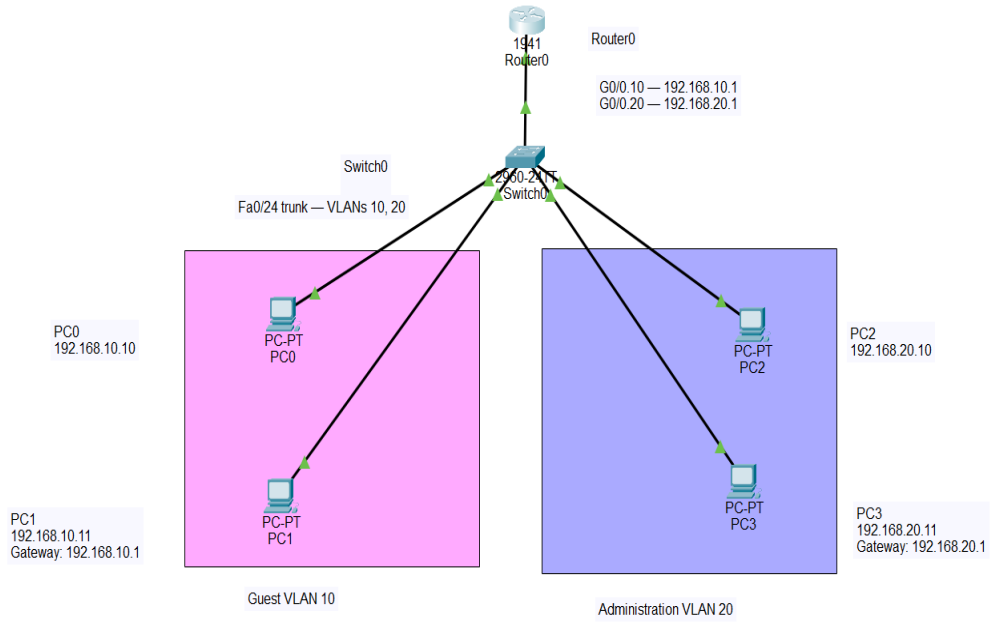


Figure 2.1: Logical topology of the hotel network.



2.4 VLAN and IP Addressing Design

Device	VLAN / Name	IP address	Subnet mask	Default gateway
PC0	10 / GUEST	192.168.10.10	255.255.255.0	192.168.10.1
PC1	10 / GUEST	192.168.10.11	255.255.255.0	192.168.10.1
PC2	20 / ADMIN	192.168.20.10	255.255.255.0	192.168.20.1
PC3	20 / ADMIN	192.168.20.11	255.255.255.0	192.168.20.1
Router G0/0.10	10 / GUEST	192.168.10.1	255.255.255.0	N/A
Router G0/0.20	20 / ADMIN	192.168.20.1	255.255.255.0	N/A

2.5 Switch Port Design

Switch port	Mode	VLAN / allowed VLANs	Connection
Fa0/1–Fa0/2	Access	VLAN 10	PC0 and PC1 (GUEST)
Fa0/9–Fa0/10	Access	VLAN 20	PC2 and PC3 (ADMIN)
Fa0/24	Trunk	10, 20	Router0 G0/0

CHAPTER 3: IMPLEMENTATION

3.1 Technologies Used

VLANs separate guest and administration devices into different Layer 2 broadcast domains.

802.1Q trunking carries VLAN 10 and VLAN 20 over one physical switch-to-router link.

Router-on-a-stick uses multiple router subinterfaces on one physical interface to provide gateways for multiple VLANs.

IPv4 addressing assigns each VLAN a separate /24 subnet.

An extended ACL filters IP traffic based on source and destination network addresses.

3.2 Configuration Steps

Step 1: Create VLANs and assign access ports on Switch0.

```
enable
configure terminal
vlan 10
 name GUEST
exit
vlan 20
 name ADMIN
exit
interface range fastEthernet 0/1-2
 switchport mode access
 switchport access vlan 10
exit
interface range fastEthernet 0/9-10
 switchport mode access
 switchport access vlan 20
exit
```

Step 2: Configure the Trunk Link

The switch port connected to the router is configured as a trunk.

```
interface fastEthernet 0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,20
 no shutdown
end
show vlan brief
show interfaces trunk
```

Step 3: Configure Router Subinterfaces

The following commands are entered on Router0.

```
enable
configure terminal
interface gigabitEthernet 0/0
 no shutdown
exit
interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
```

Step 4: Configure the Extended ACL

The ACL blocks IP traffic from the Guest subnet to the Administration subnet.

```
ip access-list extended GUEST_FILTER
 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 permit ip any any
exit
interface gigabitEthernet 0/0.10
 ip access-group GUEST_FILTER in
end
show ip interface brief
show access-lists
```

Step 5: Configure IP Addresses on PCs

Each PC is assigned an IP address, subnet mask, and default gateway.

PC	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.20.10	255.255.255.0	192.168.20.1
PC3	192.168.20.11	255.255.255.0	192.168.20.1

In Packet Tracer, open each PC, select Desktop → IP Configuration, and enter the corresponding values.

3.3 Screenshots and Outputs

A. VLAN Verification

```
Switch>show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                 Fa0/7, Fa0/8, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Gig0/1
                                                 Gig0/2
10   GUEST                             active    Fa0/1, Fa0/2
20   ADMIN                             active    Fa0/9, Fa0/10
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch>
```

B. Trunk Verification

```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      10,20

Port        Vlans allowed and active in management domain
Fa0/24      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      10,20

Switch>
```

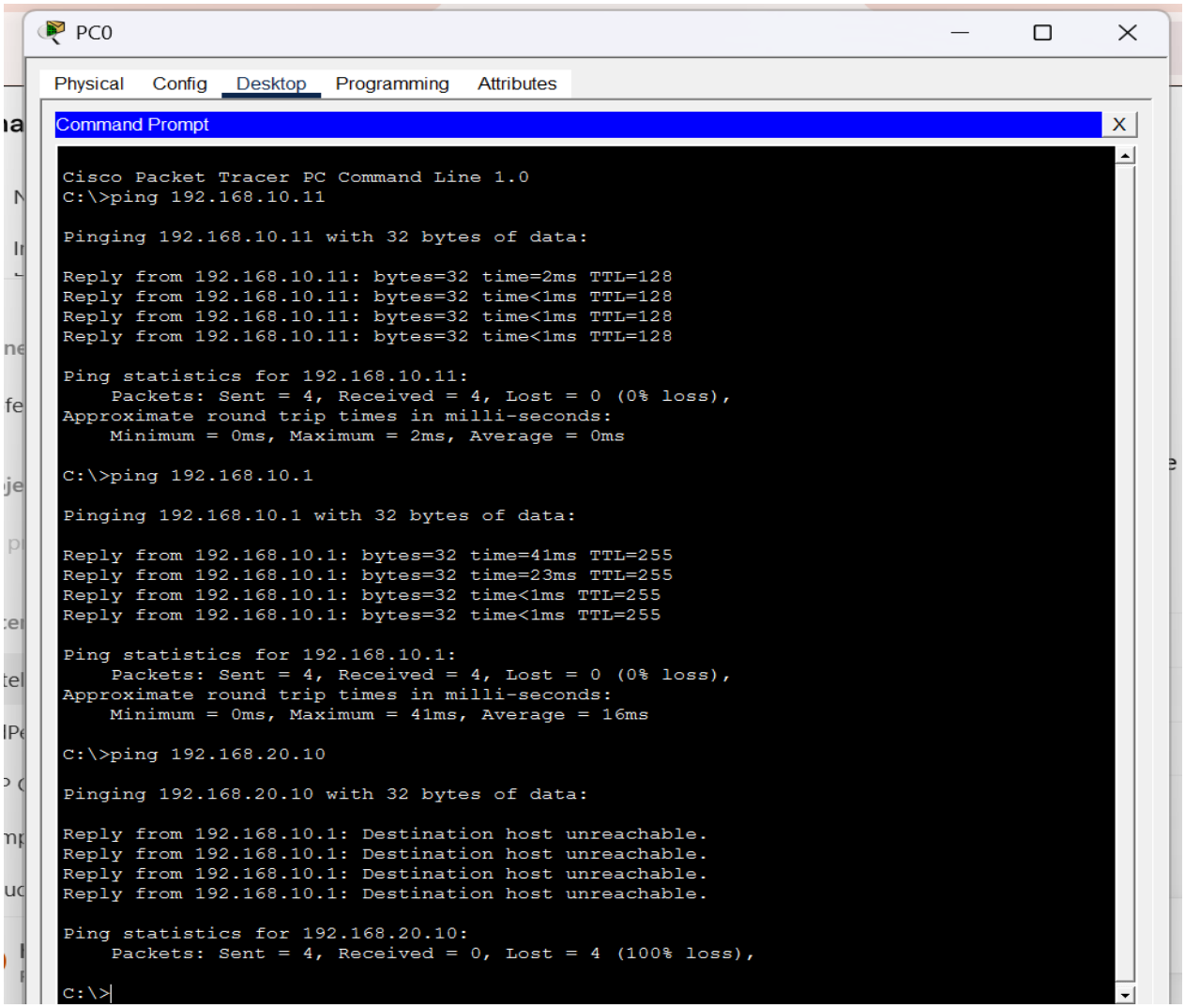
C. Router Interface Verification

```
Router>show ip  interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

D. Connectivity Testing

The following tests were performed in the simulated network.

Test	Observed result	Interpretation
PC0 → 192.168.10.1	Successful; 0% loss	Guest PC reaches its gateway.
PC0 → 192.168.10.11	Successful; 0% loss	Guest PCs communicate within VLAN 10.
PC2 → 192.168.20.1	Successful; 0% loss	Administration PC reaches its gateway.
PC2 → 192.168.20.11	Successful; 0% loss	Administration PCs communicate within VLAN 20.
PC0 → 192.168.20.10	Failed; 100% loss	Guest-to-Administration traffic is blocked.



3.4 Results and Security Analysis

The connectivity tests demonstrate that each VLAN can communicate with its own gateway and that PCs within the same VLAN can communicate with each other. The Guest-to-Administration ping failed, which is consistent with the configured ACL.

The ACL is applied inbound on the Guest router subinterface. It denies IP traffic from 192.168.10.0/24 to 192.168.20.0/24. This provides a basic example of how access-control policies can restrict communication between network segments.

However, the current configuration does not block Administration-to-Guest traffic. Also, internet access, DHCP, NAT, and wireless connectivity are not implemented. These limitations are considered possible improvements for future development.

CHAPTER 4: CONCLUSION AND FUTURE SCOPE

4.1 Conclusion

The Hotel Network Design project was implemented and tested in Cisco Packet Tracer. VLAN 10 (GUEST) and VLAN 20 (ADMIN) were created, and the PCs were assigned to the intended access ports and IP subnets.

A trunk link was configured between the switch and router, while router-on-a-stick subinterfaces provided default gateways for both VLANs. An extended ACL named GUEST_FILTER was applied inbound on the Guest subinterface to deny IP traffic from the Guest subnet to the Administration subnet.

The reported tests showed successful same-VLAN and gateway connectivity, while a Guest-to-Administration ping was blocked. The project provided practical understanding of VLAN segmentation, trunking, router subinterfaces, IPv4 addressing, ACLs, and network verification.

Overall, the project demonstrates a basic method for separating guest and administration devices in a simulated hotel network. It also highlights the importance of testing network configurations and checking access-control rules to ensure that they behave as intended.

4.2 Future Enhancements

The project can be improved in the following ways:

1. Dynamic IP Addressing: Configure DHCP pools so that PCs automatically receive IP addresses, subnet masks, and default gateways.
2. Wireless Connectivity: Add wireless access points and create a separate wireless guest network.
3. Internet Connectivity: Connect the network to an internet service and configure Network Address Translation (NAT), if required.
4. Improved Access Control: Create and test ACL rules for both traffic directions according to the hotel's security requirements.
5. Hotel Server Integration: Add servers for hotel management services and apply suitable access restrictions.
6. Network Monitoring: Introduce logging and monitoring to help administrators identify connectivity issues and suspicious traffic.

7. Enhanced Security: Implement additional security measures based on the needs of a real hotel environment.

These enhancements would make the design more realistic and suitable for a larger network.

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